GEOAI 04 change report: Madhya Pradesh (Bhopal lake & fringe)

Totals

Raw detections: 3649
Real changes after filters: 47
Area changed: 8.09 ha
Cropland affected: 0.12 ha (upper bound)
Built-up land affected: 6.31 ha
critical: 0
high: 2
medium: 7
low: 38

Open this result: /scene/madhya-pradesh-encroachment

Top 10 changes

#	Tier	Type	Area (ha)	Conf.	Reason
1	high	built_up_gain	0.73	0.974	new construction / built-up: built-up index fell (-0.09), vegetation index fell
2	high	built_up_gain	0.75	0.971	new construction / built-up: built-up index fell (-0.07), vegetation index fell
3	medium	built_up_gain	0.545	0.971	new construction / built-up: built-up index fell (-0.08), vegetation index fell
4	medium	built_up_gain	0.435	0.973	new construction / built-up: built-up index fell (-0.25), vegetation index fell
5	medium	built_up_gain	0.425	0.968	new construction / built-up: built-up index fell (-0.12), SWIR/blue ratio (paved
6	medium	built_up_gain	0.3	0.977	new construction / built-up: built-up index fell (-0.27), water index rose (+0.1
7	medium	built_up_gain	0.3	0.954	new construction / built-up: built-up index rose (+0.15), vegetation index rose
8	medium	built_up_gain	0.2	0.979	new construction / built-up: built-up index fell (-0.15), SWIR/blue ratio (paved
9	medium	built_up_gain	0.18	0.981	new construction / built-up: built-up index fell (-0.15), vegetation index fell
10	low	built_up_gain	0.175	0.969	new construction / built-up: built-up index fell (-0.18), vegetation index fell

Sources and limits

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0
- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Rivers, sandbars and lake edges move between dates and can be flagged as change; bare soil in dry regions can look like construction. Check flagged areas near water.
- Land cover is ESA WorldCover 2021, newer than the before image (2019): land already cleared or under construction can be mislabelled as cropland, so treat cropland as an upper bound
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero